

Introductory Thermodynamics Shared Reference

For the course R750:315:01

Yellow background colors or highlights represent equations or concepts that we are expected to know very well (most likely memorize).

Textbook

[Download the PDF](#)

Throughout the reference, there are page numbers that refer to this PDF's page numbers, not the book's page numbers.

Constants

Universal Gas Constant $R = 0.08206 \frac{L \cdot atm}{mol \cdot K}$ $= 8.314 \frac{J}{mol \cdot K}$	Boltzmann Constant $k = 1.381 \cdot 10^{-23} \frac{J}{K}$ $= 8.617 \cdot 10^{-5} \frac{eV}{K}$
Avogadro's Number $N_A = 6.022 \cdot 10^{23} mol^{-1}$	Planck's Constant $h = 6.626 \cdot 10^{-34} J \cdot s$ $= 4.136 \cdot 10^{-15} eV \cdot s$
Bohr Magnetron $\mu_B \equiv \frac{eh}{4\pi m_e}$ $= 9.274 \cdot 10^{-24} \frac{J}{T}$ $= 5.788 \cdot 10^{-5} \frac{eV}{T}$ Page 114 of PDF	Speed of Light $c = 299,792,458 m/s$ $\approx 3 \cdot 10^8 m/s$

Basics

Pressure, Force, and Area $P = \frac{F}{A}$ where: • P is pressure • F is force • A is area Often in the units of: • $\frac{N}{m^2} = \frac{kg}{m \cdot s^2}$ • $1 \text{ atm} = 760 \text{ torr} = 760 \text{ mmHg}$ • Pa (SI unit)	$nR = Nk$ $nR = Nk$ $n = \frac{N}{N_A}$ $k = \frac{R}{N_A}$ where: • n is the # of moles of particles • R is the Universal Gas Constant • N is the # of particles • k is the Boltzmann Constant • N_A is Avogadro's Number
Heat Capacity $C = \frac{Q}{\Delta T}$ Units: $\frac{J}{K}$ or $\frac{J}{^\circ C}$ where: • C is the heat capacity • Q is heat • ΔT is change in temperature This equation only holds when there is NO phase transition, in which case $\Delta T = 0$.	Specific Heat Capacity $c = \frac{C}{m}$ Units: usually $\frac{J}{g \cdot ^\circ C}$ where: • c (lowercase) is the specific heat capacity • C (uppercase) is the heat capacity • m is mass
Small, Large, and Very Large Numbers Small numbers are the ones we know and most often work with: 6, 90, 1005 Large numbers are made by exponentiating small numbers. A key property of them is that you can add small numbers to them without really changing them:	

$10^{23} + 50 = 10^{23} \text{ (basically)}$ Very large numbers are made by exponentiating large numbers. From the property of large numbers, multiplying a very large number by a large number basically does not change it: $10^{10^{23}} \cdot 10^{23} = 10^{(10^{23}+23)} = 10^{10^{23}} \text{ (basically)}$ You can also deal with such very large numbers by taking their logarithms (essentially undoing the power of exponentiation). Then, you can exponentiate back to get the final answer. This makes handling very large numbers easier. Page 73 of PDF	
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Terminology

Microstate	Represents the state of each particle in a system. (e.g. flipping 3 fair coins: knowing exactly <i>which</i> coin is H (heads) or T (tails))
Macrostate	Represents the overall state of the system. (e.g. flipping 3 fair coins: knowing that there were 2 H, but NOT <i>which</i> coins were H) If you know the microstate of a system, you can know the macrostate. The reverse is NOT true. <i>micro</i> → <i>macro</i>
Multiplicity	The number of microstates corresponding to a macrostate. Represented by Ω.

	(e.g. flipping 2 fair coins: 3 microstates correspond to the macrostate of 2 H).
Position Space	The regular space we know (the 3D world). Related to regular volume V . Section 2.5, Page 77 of PDF
Momentum space	An imaginary space where the axes are p_x , p_y , and p_z . Each point in this space represents a momentum vector for a particle. Related to momentum volume V_p . Section 2.5, Page 77 of PDF
Reversible	Describes processes that do NOT change the entropy of the universe. No mechanical process is perfectly reversible, but some get close. Usually changing the conditions infinitesimally is conducive to reversibility, e.g.: • Quasistatic (very slow) compression • Slow heat flow Section 2.6, Page 91 of PDF
Irreversible	Describes processes that create new entropy (increase in total entropy). Section 2.6, Page 91 of PDF
Residual Entropy	Entropy that remains as temperature approaches absolute zero. One of the three ways residual entropy are: • Different molecular orientations in a crystal requiring very little energy • Entropy of mixing (usually regarding isotopes) • Nuclear spin alignments (parallel or antiparallel to neighbors results in zero entropy, but this almost never happens) Section 3.2, Page 103-104 of PDF
Isentropic	A process that does NOT change a system's entropy . Usually such a process is quasistatic and adiabatic . Section 3.4, Page 121 of PDF

Types of Equilibrium	When two or more systems are in the following types of equilibrium, the following parameters are the same: 1. Thermal equilibrium - temperatures are the same 2. Mechanical equilibrium - pressures are the same 3. Diffusive equilibrium - chemical potentials are the same Section 3.5, Page 124, 125 of PDF
Heat Engine	<i>heat</i> $\rightarrow$ <i>engine</i> $\rightarrow$ <i>work</i> Section 4.1, Page 131 of PDF
Reservoir	Anything that is so large that its temperature does NOT change with heat flow. Section 4.1, Page 131 of PDF
Thermodynamic Potentials	Collectively refers to U (thermal energy), H (enthalpy), F (Helmholtz free energy), and G (Gibbs free energy).
Extensive	An extensive property depends on the amount of material (e.g. volume is extensive because there is more of it with more stuff). Section 5.2, Page 172 of PDF
Intensive	An intensive property is independent of the amount of material (e.g. density is the same no matter how large the material is). Section 5.2, Page 172 of PDF
Equation of State	Any proposed relationship between P, V, and T (such as the ideal gas law or the van der Waals equation). Section 5.3, Page 189 of PDF
Bosons	Particles that can share a state with another particle of the same species. They have integer spin (0, 1, 2, 3, ...). Examples: photons, pions, He-4 Section 7.2, Page 272 of PDF
Fermions	Particles that CANNOT share a state with another particle of the same species. They have half-integer spin (1/2, 3/2, ...). Examples: electrons, protons, neutrons, neutrinos Section 7.2, Page 272 of PDF
Occupancy	The occupancy of a state is the average number of particles in that state. Section 7.2, Page 275 of PDF
(Fermi) Degenerate	All states below the fermi energy level are occupied while all the states above it are unoccupied. (This is completely different than degenerate

	energy levels, whose energies are the same). Section 7.3, Page 281 of PDF
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Large-Scale Concepts

Fundamental Assumption of Statistical Mechanics In an isolated system at thermal equilibrium, all accessible microstates are equally probable. Page 69 of PDF	Principle of Detailed Balance A system can just as easily go from state X to state Y as from state Y to state X (reversibility). Page 69 of PDF
Thermodynamic Limit The limit where the system becomes infinitely large, causing an increasingly sharper peak of its multiplicity for each macrostate. Quite quickly, it becomes so sharp that fluctuations from the most likely macrostate (the middle for 2 equally-sized systems) basically never occur. Page 78 of PDF	Second Law of Thermodynamics “Any large system in equilibrium will be found in the macrostate with the greatest multiplicity [or entropy] (aside from fluctuations that are normally too small to measure).” Multiplicity and entropy tend to increase. Page 83 of PDF
Third Law of Thermodynamics As temperature approaches zero, the entropy reaches a constant value. Ideally, $\Omega = 1$ and $S = 0$, but this often does not happen due to residual entropy. It is also written as: $C_V \rightarrow 0 \text{ as } T \rightarrow 0$ To ensure that the entropy remains finite and positive. Page 102-104 of PDF	Pauli Exclusion Principle No two fermions can occupy the same state. Page 272 of PDF

Chapter 1

1.2 Ideal Gas Law (Form nR)

$$PV = nRT$$

where:

- P is pressure
- V is volume
- n is the # of **moles** gas particles
- R is the [Universal Gas Constant](#)
- T is the temperature in **Kelvin**

★ 1.2 Ideal Gas Law (Form Nk)

$$PV = NkT$$

where:

- P is pressure
- V is volume
- N is the # of gas particles
- k is the [Boltzmann constant](#)
- T is the temperature in **Kelvin**

This is the **preferred form** for this course.

1.3 Degrees of Freedom

f	Structure	f	Structure
3	Monatomic molecule <i>all translational</i>	6	Solid <i>6 vibrational</i> • 3 kinetic • 3 potential
5	Diatomic molecule at room temperature <i>3 translational</i> <i>2 rotational (spin does NOT count)</i>	7	Diatomic molecule at high temperature <i>3 translational</i> <i>2 rotational (spin does NOT count)</i> <i>2 vibrational</i> • 1 kinetic • 1 potential

★ 1.3 Thermal Energy

$$U_{thermal} = N \cdot f \cdot \frac{1}{2}kT$$

where:

- $U_{thermal}$ is the thermal energy
- N is the # of particles
- f is the degree of freedom
- k is the [Boltzmann Constant](#)

★ 1.4 First Law of Thermodynamics

$$\Delta U_{thermal} = Q + W$$

where:

- $\Delta U_{thermal}$ is the change in thermal energy
- Q is heat
- W is work

• T is the temperature in Kelvin	Q is spontaneous heat transfer (leaving a coffee out in the cold, etc.). Does NOT include things like microwave heating, which is really W. W is every other form of a system gaining or losing thermal energy (rubbing, microwave heating, etc.).
1.5 Compression Work $W = -P\Delta V = - \int_{V_i}^{V_f} P(V) dV$ where: • W is work • P is pressure • ΔV is the change in volume If the gas expands, the work is negative (the gas does work on the surroundings). If the gas compresses, the work is positive (surroundings do work on the gas).	1.5 Quasistatic Compression When you compress the gas so slowly that it remains in equilibrium, Newton's 3rd law can be used to write: $F_{\text{applied}} = F_{\text{gas}} = PA$
1.5 Non-Quasistatic Compression Very fast compression or you push very hard. Usually... $F_{\text{applied}} > PA$ and $W_{n\text{-quasi}} > W_{\text{quasi}}$	where: • F_{applied} is the force by the piston on the gas • F_{gas} is the force by the gas on the piston • P is the gas pressure • A is the area of the piston • W is work
★ 1.5 Isothermal Compression/Expansion $Q \neq 0$ $\Delta U = 0$	★ 1.5 Adiabatic Compression/Expansion $Q = 0$ $\Delta U = W$

so...

$$Q = -W$$

$$Q = NkT \ln \frac{V_f}{V_i}$$

where:

- Q is heat
- ΔU is the change in internal energy
- W is work
- N is the # of particles
- k is the [Boltzmann Constant](#)
- T is temperature (constant in this case)
- V_f and V_i are final and initial volumes

You perform the compression or expansion so **slowly** that the generated heat escapes and the system is always in (thermal) equilibrium with the surroundings.

You perform the compression or expansion so **fast** that the system cannot lose or gain heat in that amount of time to accommodate the work done on it.

$$VT^{f/2} = \text{constant}$$

$$PV^\gamma = \text{constant}$$

$$\gamma = 1 + \frac{2}{f} \text{ (adiabatic exponent)}$$

where:

- Q is heat
- ΔU is the change in internal energy
- W is work
- T is temperature (constant in this case)
- V is volume
- P is pressure
- f is degrees of freedom

1.6 Constant Volume Heat Capacity

$$C_V = \left[\frac{\partial U}{\partial T} \right]_V$$

where:

- C_V is the “energy capacity” (heat capacity at constant volume)
- $\left[\frac{\partial U}{\partial T} \right]_V$ is the derivative of thermal energy with respect to temperature (at constant volume)

No compression work is done. This formula is derived from [★1.4 First Law of Thermodynamics](#).

1.6 Constant Pressure Heat Capacity

$$C_P = \left[\frac{\partial U}{\partial T} \right]_P + P \left[\frac{\partial V}{\partial T} \right]_P = \left[\frac{\partial H}{\partial T} \right]_P$$

where:

- C_P is the heat capacity at constant pressure
- $\frac{\partial U}{\partial T}$ is the derivative of thermal energy with respect to temperature (at constant pressure)
- P is pressure (constant)
- $\left[\frac{\partial V}{\partial T} \right]_P$ is the derivative of volume with respect to temperature (at constant pressure)
- $\left[\frac{\partial H}{\partial T} \right]_P$ is the derivative of enthalpy with respect to temperature (at constant pressure)

1.6 Ideal Gas Constant Volume Heat Capacity $C_V = \frac{f}{2} Nk = \frac{f}{2} nR$ where: • C_V is the heat capacity at constant volume • f is degrees of freedom • N is the # of gas particles • k is the Boltzmann constant • n is the # of moles gas particles • R is the Universal Gas Constant This is only for the ideal gasses because we substitute $V = \frac{NkT}{P}$ and $P = \frac{NkT}{V}$.	1.6 Ideal Gas Constant Pressure Heat Capacity $C_P = C_V + Nk = C_V + nR$ where: • C_P is the heat capacity at constant pressure • C_V is the heat capacity at constant volume • N is the # of gas particles • k is the Boltzmann constant • n is the # of moles gas particles • R is the Universal Gas Constant This is only for the ideal gasses because we substitute $V = \frac{NkT}{P}$.
1.6 Enthalpy $H \equiv U + PV$ where: • H is enthalpy • U is thermal energy • P is pressure • V is volume	1.6 Latent Heat $l \equiv \frac{L}{m} = \frac{Q}{m}$ where: • l is specific latent heat • L is latent heat • m is mass • Q is heat

Chapter 2

2.1 Multiplicity of Two-State Systems $\Omega(N, n) = \frac{N!}{n! \cdot (N - n)!} = \binom{N}{n}$ where: • Ω is the multiplicity of microstates in a macrostate • N is the number of objects to choose from	2.2 Multiplicity of Einstein Solids An Einstein solid consists of N identical quantum harmonic oscillators (with quantized energy), which can each oscillate in 3 directions. Thus, there are $N/3$ atoms. $\Omega(N, q) = \binom{q + N - 1}{q} = \frac{(q + N - 1)!}{q! (N - 1)!}$
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n is the number of objects chosen This equation is for two-state systems, such as flipping a coin (one of two possible states: H or T). 2.1 Probability of a Macrostate $p = \frac{\Omega(\text{macro})}{\Omega(\text{all macros})}$ where: $\Omega(\text{macro})$ is the multiplicity of the desired macrostate $\Omega(\text{all macros})$ is the total multiplicity of the system (all possible microstates)	where: Ω is the multiplicity of a microstate in a macrostate N is the number of quantum harmonic oscillators q is the number of energy units In the following illustration, there are $N = 4$ quantum harmonic oscillators: $q(1) = 1$, $q(2) = 3$, $q(3) = 0$, $q(4) = 4$.  $q + N - 1$ represents the total number of symbols, with q energy units (dots), and $N - 1$ partitions (lines).
2.3 Multiplicity of 2 Einstein Solids for a Certain Macrostate $\Omega_{\text{total, macro}} = \Omega_A \cdot \Omega_B$ where: $\Omega_{\text{total, macro}}$ is the total multiplicity of a 2-solid system for a certain macrostate Ω_A is the multiplicity of solid A Ω_B is the multiplicity of solid B Page 57 of PDF	2.3 Total Multiplicity of 2 Einstein Solids for All Macrostates $\Omega(N_A, q_A, N_B, q_B) = \binom{N_A + N_B + q_A + q_B - 1}{q_A + q_B}$ where: $\Omega(N_A, q_A, N_B, q_B)$ is the grand total multiplicity of all macrostates for 2 Einstein solids N_A and N_B are the number of oscillators in solid A and B, respectively q_A and q_B are the number of energy units in solid A and B, respectively This is also the sum of all multiplicities for each macrostate. Page 57 of PDF
2.4 Stirling's Approximation $N! \approx N^N e^{-N} \sqrt{2\pi N}$	2.4 Multiplicity of Large Systems (High Temperature Limit)

$$\approx N^N e^{-N} \text{ (for large } N\text{)}$$

$$\ln N! \approx N \ln N - N$$

where:

- N is a number (yep)

The second line is the approximation we will use in class, without the factor $\sqrt{2\pi N}$.

The third line is the natural log of both sides, useful in some cases.

$N!$ is approximately N^N , but the latter grossly overestimates the former. However, the ratio of the former to the latter turns out to be around e^{-N} . This is a correcting factor. Finally, $\sqrt{2\pi N}$ is for even more correction.

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$$\Omega(N, q) \approx \left(\frac{eq}{N}\right)^N$$

$$(q \gg N)$$

where:

- Ω is the multiplicity of a system (# of microstates for the given macrostate)
- q is a large # of energy units (much larger than N , though)
- N is a large # of oscillators

Remember, this is for the high-temperature limit, which means $q \gg N$.

If you increase N or q even by a little bit, Ω grows by a *lot* because of exponentiation.

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2.4 Width of the Multiplicity of Large Systems

$$2x = \frac{q}{\sqrt{N}}$$

$$(q \gg N)$$

where:

- $2x$ is the width (whose bounds are where the peak falls off by $1/e$)
- q is the number of energy units
- N is the number of oscillators

2.5 Volume in Momentum Space

One particle:

$$V_p = p_x^2 + p_y^2 + p_z^2 = 2mU$$

Two particles:

$$= p_{1x}^2 + p_{1y}^2 + p_{1z}^2 + p_{2x}^2 + p_{2y}^2 + p_{2z}^2$$

$$= 2mU$$

where:

- V_p is the volume (or surface area) of a momentum sphere
- p_x , p_y , and p_z are momentum in each respective axis, and the numbers in the subscripts represent the particle in question
- m is mass

	• U is the kinetic energy Note: • The momentum sphere volume and surface area are considered interchangeable, though the units are always $(kg \cdot \frac{m}{s})^3$. • This is similar to the equation of a sphere in Calculus 3, so the radius of a momentum sphere is $\sqrt{2mU}$. • Two particles create a 6-dimensional hypersphere, still with the radius of $\sqrt{2mU}$ Page 77, 79 of PDF
2.5 Multiplicity of One Gas Particle $\Omega_1 = \frac{V \cdot V_p}{h^3}$ where • Ω_1 is the multiplicity of a single gas molecule in a fixed volume • V is some fixed volume (of position space) • V_p is the volume of momentum space • h is Planck's Constant Note: • While V_p is volume in momentum space, it is the surface area of a sphere in such space. We can usually add a bit of thickness to that surface to make it into volume, but the units of V_p are $(kg \cdot \frac{m}{s})^3$. (Remember, multiplicity has NO units.) • h is cubed due to this being in 3 dimensions. Page 79 of PDF	2.5 Multiplicity of Two Gas Particles Two distinguishable particles: $\Omega_2 = \frac{V^2}{h^6} \cdot (\text{area of } p \text{ sphere})$ Two indistinguishable particles: $\Omega_2 = \frac{1}{2} \frac{V^2}{h^6} \cdot (\text{area of } p \text{ sphere})$ where: • Ω_2 is the multiplicity of two gas molecule in a fixed volume • <i>(area of p sphere)</i> is the surface area (aka volume) of a momentum hypersphere (the expression is complicated, so it is represented conceptually) Note: • The 1/2 coefficient is added for indistinguishable particles to account for duplicate counting (whether particle 1 is in a certain state or

particle 2 is in that same state doesn't matter, they're identical particles).

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2.5 Surface Area of a d-Dimensional Hypersphere

$$A = \frac{2\pi^{d/2}}{(d/2 - 1)!} r^{d-1}$$

where:

- d is the dimension in question (3 for 3 dimensions, 2 for 2 dimensions, etc.)

Note:

- $(1/2)! = \sqrt{\pi}/2$ (crazy!)

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2.5 Multiplicity of Any Number of Monatomic Gas Particles

$$\Omega_N = \frac{1}{N!} \frac{V^N}{h^{3N}} A \quad (1)$$

$$A = \frac{2\pi^{3N/2}}{(3\pi/2 - 1)!} (2mU)^{(3N-1)/2} \quad (2)$$

$$\approx \frac{\pi^{3N/2}}{(3\pi/2)!} (2mU)^{3N/2} \quad (3)$$

$$\Omega(U, V, N) = f(N) V^N U^{3N/2} \quad (4)$$

where:

- N is the number of particles
- V is volume
- h is [Planck's Constant](#)
- A is the surface area of the momentum hypersphere
- m is mass
- U is kinetic energy
- f is a complicated function of N

Note:

- Equation 4 is just a simplification of equation 1 with A substituted with equation 3 (the approximation for A). We can see that the multiplicity depends on the kinetic energy, volume, and number of particles.
- f contains all the coefficients and other parts of the equation separate from V and U

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2.5 Multiplicity of Two Interacting Ideal Gasses

$$\Omega_{total} = [f(N)]^2 (V_A V_B)^N (U_A U_B)^{3N/2}$$

where:

- f is a complicated function of N
- N is the number of particles
- V_A and V_B are the volumes of ideal gasses A and B, respectively
- U_A and U_B are the kinetic energies of ideal gasses A and B, respectively

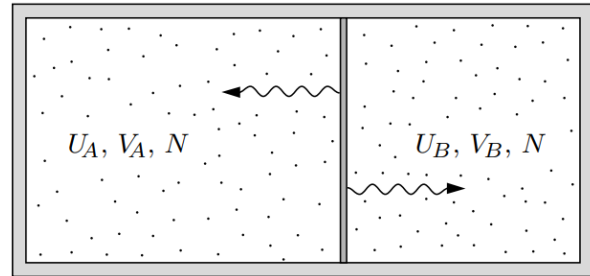
Note:

- This is just the multiplication of [2.5 Multiplicity of Any Number of Monatomic Gas Particles](#)

2.5 Widths of Probable Macrostates of Two Interacting Ideal Gasses

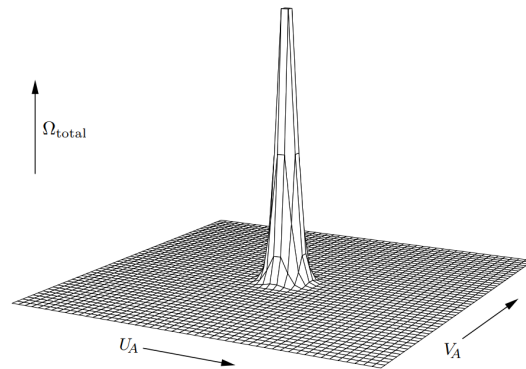
Energy exchange only:

$$\text{width of peak} = \frac{U_{total}}{\sqrt{3N/2}}$$



Energy and volume exchange:

$$\text{width of peak} = \frac{V_{total}}{\sqrt{N}}$$



where:

- U_{total} is the total kinetic energy (fixed)
- V_{total} is the total volume (fixed)
- N is the number of particles

Note:

- The width has the same definition as before: where the multiplicity drops off by $1/e$.

2.6 Entropy

$$S \equiv k \ln \Omega$$

where:

- S is entropy
- k is the [Boltzmann Constant](#)
- Ω is the multiplicity

Whenever Ω increases (the number of ways to arrange particles and energy), S increases.

These cases include, but aren't limited to:

- Adding more particles
- Allowing a substance to move into a larger volume
- Mixing once-separate substances
- Breaking molecules apart into smaller ones

The peak in the graph of entropy against, for instance, energy or volume is not as sharp, but still, the most likely macrostate wins by a lot.

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2.6 Total Entropy

$$S_{total} = k \ln(\Omega_A \Omega_B)$$

where:

- k is the [Boltzmann Constant](#)
- Ω_A and Ω_B is the multiplicity of system A and system B

Please see the assumption made (though valid) in the page listed below.

In short, this isn't the same for interacting systems, just systems whose macrostates are specified separately. However, in reality the difference is negligible.

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2.6 Sackur-Tetrode Equation (Entropy of an Ideal Gas)

$$S = Nk \left[\ln \left(\frac{V}{N} \left(\frac{4\pi m U}{3N h^2} \right)^{3/2} \right) + \frac{5}{2} \right]$$

where:

- N is the number of particles
- k is the [Boltzmann Constant](#)
- V is volume
- m is particle mass
- U is energy
- h is [Planck's Constant](#)

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2.6 Entropy Change vs Volume Change

$$\Delta S = Nk \ln \frac{V_f}{V_i}$$

(U, N fixed)

where:

- N is the number of particles
- k is the [Boltzmann Constant](#)
- V_f and V_i is final and initial volume, respectively

This is the simplest relationship derived from the [Sackur-Tetrode Equation](#). Remember, energy and number of particles are required to be **constant**.

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2.6 Entropy Change due to Heat $\Delta S = \frac{Q}{T}$ where: • Q is heat • T is temperature This holds true for any quasistatic process. Page 88 of PDF	2.6 Entropy of Mixing $\Delta S = \Delta S_A + \Delta S_B$ where: • ΔS_A and ΔS_B are changes in entropy of gasses A and B, respectively This equation derives from Entropy Change vs Volume Change. If the gasses end up in the same container, the volume change of gas A leads to ΔS_A, and similarly for gas B. Then you simply add them. Page 89 of PDF
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Chapter 3

3.1 Thermal Equilibrium $\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B}$ where: • S_A and S_B are the entropies of Einstein Solid A and B, respectively • U_A and U_B are the thermal energies of Einstein Solid A and B, respectively Page 96 of PDF	3.1 Temperature Definition $T \equiv \left(\frac{\partial S}{\partial U} \right)^{-1}_{N, V}$ (N, V fixed) where: • S is the entropy • U is the thermal energy • N is number of particles • V is volume Often convenient to represent in terms of ϵ/k, which eventually becomes Kelvin. Page 97 of PDF
3.1 Energy Units in Einstein Solids $U = q\epsilon$ $\epsilon = hf$ where:	3.2 Game Plan: Multiplicity to Heat Capacity 1. Derive Ω (hardest part). 2. Calculate $S = k \ln \Omega$.

- U is the thermal energy
- q is the number of energy units
- ϵ is the amount of energy in one unit
- h is [Planck's Constant](#)
- f is frequency (for quantum harmonic oscillators)

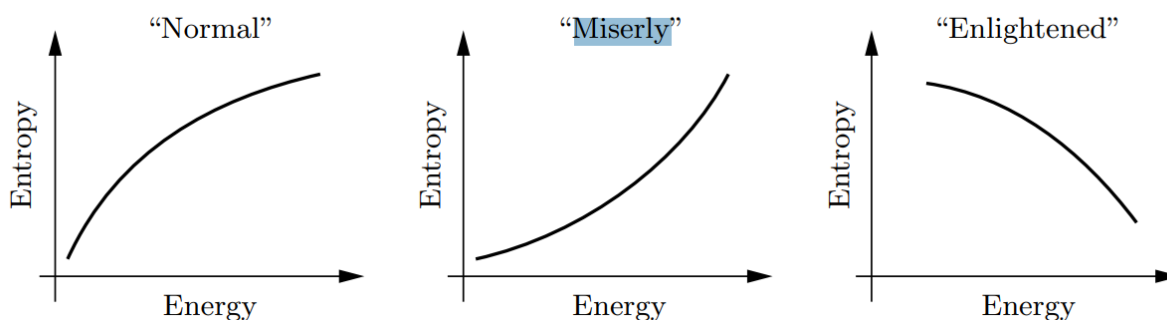
You can do calculations in terms of ϵ for simplicity, e.g. representing temperature in terms of ϵ/k .

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3. Calculate $T = \left(\frac{\partial S}{\partial U}\right)^{-1}_{N,V}$.
4. Solve for U in terms of T .
5. Calculate $\left(\frac{\partial U}{\partial T}\right)_{N,V} = C_V$.

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3.2 Normal, Miserly, and Enlightened System



- **Normal**: Einstein solids, ideal gasses
- **Miserly**: stars
- **Enlightened**: 2-state paramagnets

3.2 Change in Entropy

$$dS = \frac{dU}{T} = \frac{Q}{T}$$

(constant volume, no work)

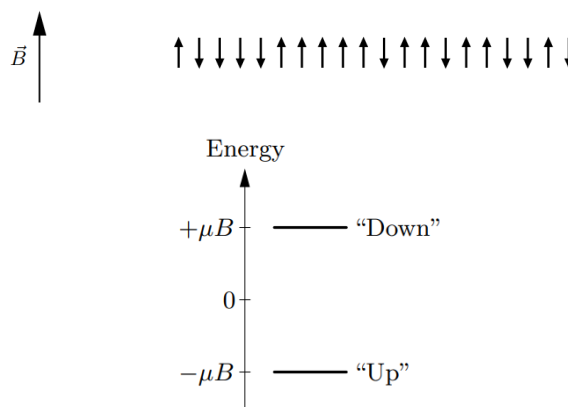
$$dS = \frac{C_V}{T} dT$$

$$\Delta S = S_f - S_i = \int_{T_i}^{T_f} \frac{C_V}{T} dT$$

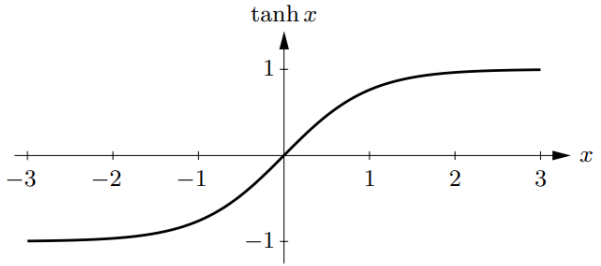
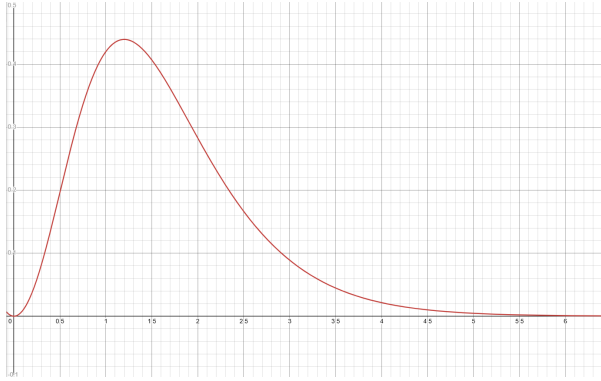
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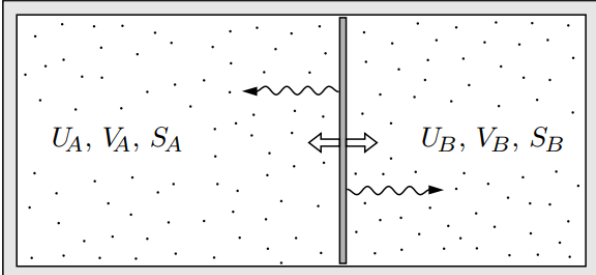
- S is entropy
- U is thermal energy
- Q is heat

3.3 2-State Paramagnets



When a dipole is aligned **with** the magnetic field, it is at the **lowest** potential energy.
When a dipole is aligned **against** the

• T is temperature • C_V is constant-volume heat capacity Page 102, 103 of PDF	magnetic field, it is at the highest potential energy. Page 107, 108 of PDF
3.3 Magnetization $M = \mu(N_{\uparrow} - N_{\downarrow}) = -\frac{U}{B}$ where: • μ is the magnetic dipole moment (positive for up dipoles) • $N_{\uparrow}$ is the number of dipoles pointing upward • $N_{\downarrow}$ is the number of dipoles pointing downward • U is the total energy • B is the magnitude of the magnetic field Page 108 of PDF	3.3 Energy of a 2-State Paramagnet $U = -N\mu B \tanh\left(\frac{\mu B}{kT}\right)$ where: • N is the total number of dipoles • μ is the magnetic dipole moment (positive for up dipoles) • B is the magnitude of the magnetic field • k is the Boltzmann Constant • T is the temperature Hyperbolic tangent function:  Page 113 of PDF
3.3 Heat Capacity of a 2-State Paramagnet $C_B = \left(\frac{\partial U}{\partial T}\right)_{N, B} = Nk \cdot \frac{(\mu B/kT)^2}{\cosh^2(\mu B/kT)}$ where: • C_B is the heat capacity at constant magnetic field • N is the total number of dipoles • k is the Boltzmann Constant • μ is the magnetic dipole moment (positive for up dipoles)	The function if the argument is just x:  The heat capacity approaches zero for both high and low temperatures.

• B is the magnitude of the magnetic field • T is the temperature	Page 114 of PDF
3.3 Curie's Law Curie's Law is a proportionality statement of $M \propto \frac{1}{T}$. Specifically, it is: $M \approx \frac{N\mu^2 B}{kT}$ (when $\mu B \ll kT$) where: • M is magnetization • N is the total number of dipoles • μ is the magnetic dipole moment (positive for up dipoles) • B is the magnitude of the magnetic field • k is the Boltzmann Constant • T is the temperature Page 114 of PDF	3.4 Mechanical Equilibrium Scenario: fixed total volume and number of particles, movable partition  $\frac{\partial S_A}{\partial V_A} = \frac{\partial S_B}{\partial V_B}$ where: • S_A and S_B are the entropies of gas A and B, respectively • V_A and V_B are the volumes of gas A and B, respectively Page 118 of PDF
3.4 Pressure, Entropy, and Volume $P = T \left(\frac{\partial S}{\partial V} \right)_{U, N}$ (mechanical equilibrium) where: • T is temperature • S is entropy • V is volume Thermal energy, U, and number of particles, N, remains constant.	★ 3.4 Thermodynamic Identity (TI) $dU = TdS - PdV$ where: • U is thermal energy • T is temperature • S is entropy • P is pressure • V is volume Memorize this equation as it can be used to derive 3.1 Temperature Definition and 3.4 Pressure, Entropy, and Volume.

Intuitively, the pressure is high when there is a large increase in entropy for a very little increase in volume. So, the system “wants” to expand.

Substituting [2.5 Multiplicity of Any Number of Monatomic Gas Particles](#) equation 4 for S yields the [1.2 Ideal Gas Law \(Form \$Nk\$ \)](#).

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3.4 Quasistatic TI

$$W = - PdV$$

so...

$$Q = TdS$$

(quasistatic)

ONLY true during **quasistatic** conditions.

3.4 Constant Pressure Change in Entropy

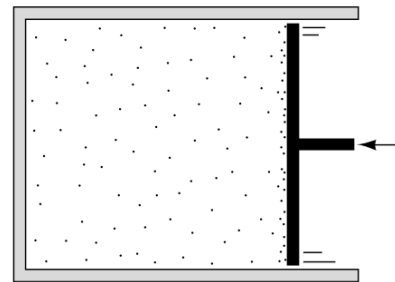
$$(\Delta S)_P = \int_{T_i}^{T_f} \frac{C_p}{T} dT$$

where

- C_p is the constant-pressure heat capacity
- T is temperature

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3.4 Nonquasistatic TI



Example: pushing a piston hard and compressing the gas faster than the velocity of the gas particles. *Extra* entropy is created:

$$dS > \frac{Q}{T}$$

because

$$|W| > |-PdV|$$

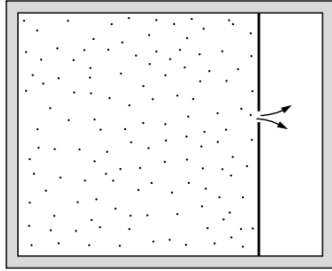
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3.4 Free Expansion TI

3.5 Chemical Potential

$$\mu \equiv -T \left(\frac{\partial S}{\partial N} \right)_{U, V} \quad (1)$$

$$\mu = \left(\frac{\partial U}{\partial N} \right)_{S, V} \quad (2)$$



$$T dS = P dV > 0$$

Entropy and volume increase without heat flow and zero change in thermal energy. ($dU = 0$ in the [★3.4 Thermodynamic Identity \(TI\)](#))

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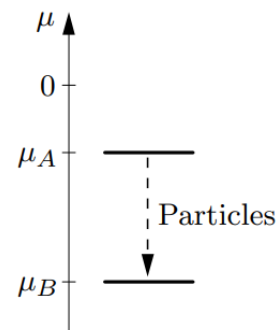
where:

- T is temperature
- S is entropy
- N is number of particles
- U is thermal energy

Note:

- Equation 1 - U and V are held constant
- Equation 2 - S and V are held constant

$\mu < 0$ always. Systems tend to decrease their chemical potential.



Equation 2 is interesting: it represents the amount of energy you need (to add or subtract) to keep entropy constant when adding or removing a particle from a system.

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Usually, increasing the density of particles increases a system's chemical potential.

Each species has its own chemical potential.

$$\mu_1, \mu_2, \dots, \mu_N$$

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3.5 Diffusive Equilibrium

$$\frac{\partial S_A}{\partial N_A} = \frac{\partial S_B}{\partial N_B}$$

Multiply both sides by $-T$

★3.5 General Thermodynamic Identity

$$dU = TdS - PdV + \mu dN$$

where:

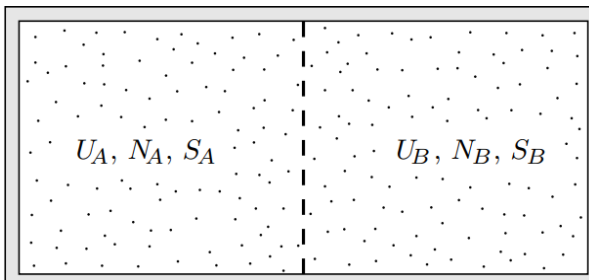
- U is thermal energy

$$\mu_A = \mu_B$$

(at equilibrium)

where:

- S_A and S_B are the entropies of gas A and B, respectively
- N_A and N_B are the number of particles of gas A and B, respectively



The system can exchange both energy and particles.

Note: each type of particle has its own chemical potential, μ_N .

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- T is temperature
- S is entropy
- P is pressure
- V is volume
- μ is chemical potential
- N is the number of particles

Note: this is for **one** kind of particle (e.g. all water vapor).

Memorizing this equation can help you derive all the partial derivatives for the different quantities (given that any two quantities are constant). For example:

1. U, V fixed $\rightarrow \mu$ (definition - eq 1)
2. S, V fixed $\rightarrow \mu$ (equation 2)

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The chemical potential part of the identity changes a bit for systems with more than one type of species:

$$dU = TdS - PdV + \sum_i \mu_i dN_i$$

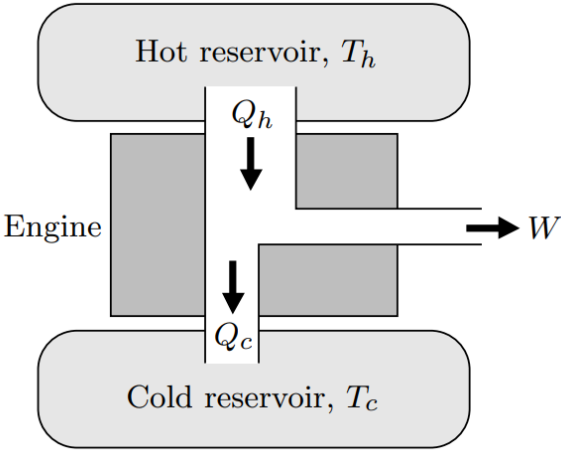
where

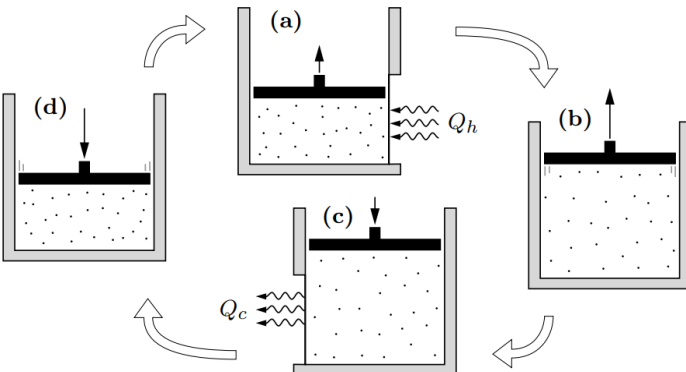
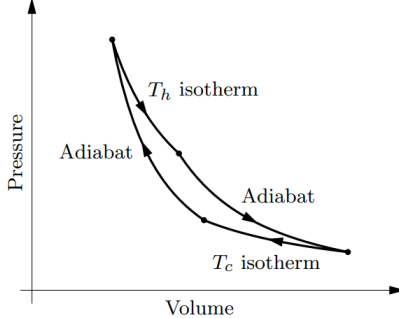
- $i = 1, 2, \dots, N$ represent the individual species

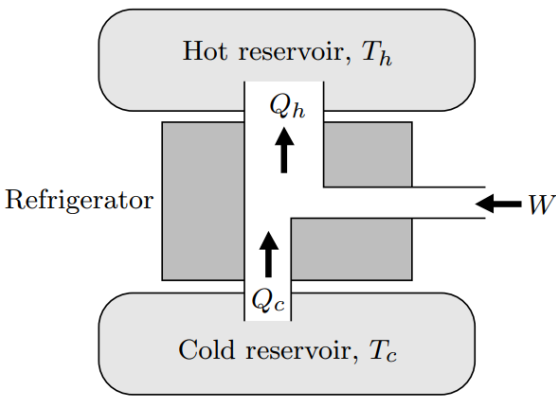
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Chapter 4

Engines and Refrigerators

	4.1 Heat Engine Efficiency $e \equiv \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h} \leq 1$ where: • W is the work output (benefit) • Q_h is the heat input (cost) (from the hot reservoir) • Q_c is the wasted heat (into the cold reservoir) Assume that $Q_h, Q_c, W > 0$. Page 132 of PDF
4.1 Maximum Heat Engine Efficiency $e \leq 1 - \frac{T_c}{T_h}$ With an equal sign, it's the best possible efficiency of a Carnot cycle. Page 132 of PDF	where: • T_h is the hot reservoir temperature • T_c is the cold reservoir temperature This factors in the Second Law of Thermodynamics.

Carnot Cycle Aims to convert heat to work <u>without increasing the entropy</u> of the gas.	
	

 Everything works in reverse from the engine (even the Carnot cycle). Though remember: it is wildly impractical due to the amount of time required for the isothermal sections.	4.2 Refrigerator Coefficient of Performance $COP = \frac{Q_c}{W} = \frac{1}{Q_h/Q_c - 1} \geq 1$ where: • Q_c is the heat absorbed (benefit) (from the cold reservoir) • W is the work input (cost) • Q_h is the waste heat released (into the hot reservoir) Usually the cold reservoir is the inside of a fridge; the hot reservoir is the kitchen; and the work is the electricity supplied. Page 137 of PDF
4.2 Maximum Refrigerator COP $COP \leq \frac{T_c}{T_h - T_c}$ Page 137 of PDF	where: • T_c is the cold reservoir temperature • T_h is the hot reservoir temperature This factors in the Second Law of Thermodynamics.

Chapter 5

5.1 Helmholtz Free Energy $F \equiv U - TS$ $\Delta F \leq W$ (at constant T) Inequality broken down: $\Delta F = W$ (quasistatic / reversible)	5.1 Gibbs Free Energy $G \equiv U - TS + PV$ $= H - TS = F + PV$ $\Delta G \leq W_{other}$ (at constant T, P) Inequality broken down: $\Delta G = W_{other}$
---	---

$\Delta F < W$ (non-quasistatic / irreversible) where: • F is Helmholtz free energy • U is thermal energy • T is temperature • S is entropy • W is <i>all</i> types of work The heat TS flows in from the environment. Page 159 of PDF	(quasistatic / reversible) $\Delta G < W_{other}$ (non-quasistatic / irreversible) where: • G is Gibbs free energy • U is thermal energy • T is temperature • S is entropy • P is pressure • V is volume • W_{other} is work that is not from the environment (e.g. W_{other} can be electrical, etc.) Page 159, 160 of PDF
5.1 Relationship Between the Thermodynamic Potentials $\begin{array}{c} \xrightarrow{-TS} \\ \begin{array}{ c c } \hline U & F \\ \hline H & G \\ \hline \end{array} \\ \downarrow +PV \end{array}$ Page 160 of PDF	5.1 Thermodynamic Identifies for H, F, and G $dH = TdS + VdP + \mu dN$ $dF = -SdT - PdV + \mu dN$ $dG = -SdT + VdP + \mu dN$ [DO ALL THE PARTIALS!] where: • T is temperature • S is entropy • V is volume • P is pressure • μ is chemical potential • N is number of particles You can set any two quantities constant and derive equations such as: $P = - \left(\frac{\partial F}{\partial V} \right)_{T,N}$ Page 166 of PDF

5.2 Change in Total Entropy

$$dS_{total} = -\frac{1}{T}dF$$

(constant T, V, N)

$$dS_{total} = -\frac{1}{T}dG$$

(constant T, P, N)

where:

- S is entropy
- T is temperature
- F is Helmholtz free energy
- G is Gibbs free energy

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5.2 Extensive and Intensive Properties**Extensive:**

Depends on the amount of stuff.

$$V, N, S, U, H, F, G, m$$

Intensive:

Doesn't depend on the amount of stuff.

$$T, P, \mu, \rho$$

Operation Rules

$ext \times int$	ext
ext / ext	int
$ext \times ext$	NEITHER
$ext \pm ext$	ext
$int \pm int$	int
$ext \pm int$	NOT ALLOWED

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5.2 Chemical Potential as Gibbs Free Energy per Particle

$$G = \sum_i^{\infty} N_i \mu_i$$

(constant T, P)

where:

- G is Gibbs free energy
- N is number of particles
- μ is chemical potential
- i represents each quantity for each type of particle

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5.2 Chemical Potential for an Ideal Gas

$$\mu(T, P) = \mu^{\circ}(T) + kT \ln(P/P^{\circ})$$

where:

- μ is chemical potential
- μ° is chemical potential at atmospheric pressure
- k is the [Boltzmann Constant](#)
- T is temperature
- P is pressure
- P° is atmospheric pressure

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5.3 Clausius-Clapeyron Relation

$$\frac{dP}{dT} = \frac{S_1 - S_2}{V_1 - V_2} = \frac{L}{T\Delta V}$$

where:

- P is pressure
- T is temperature
- S is entropy (S_i for each phase)
- V is volume (V_i for each phase)
- L is latent heat (not specific)

The slope of the boundary between two phases. Applies to *any* phase boundary on a P-T graph.

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5.3 Van Der Waals Equation

$$(P + a\frac{N^2}{V^2})(V - Nb) = NkT$$

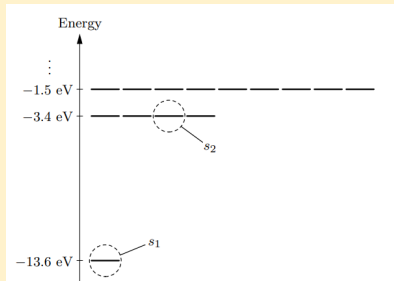
Units of a : $J \cdot m^3$

where:

- P is pressure
- N is number of particles
- V is volume
- k is the [Boltzmann Constant](#)
- T is temperature
- a, b are constants based on the gas

$\frac{aN^2}{V}$ is the potential energy. This equation is an improvement on the ideal gas model but is still an approximation.

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Chapter 6**6.1 Hydrogen Atom Energy Levels**

We need to remember that the ground state of the hydrogen atom is -13.6 eV . The degeneracy for an energy level n is n^2 .

6.1 Boltzmann / Canonical Distribution

$$p(s) = \frac{1}{Z} e^{-E(s)/kT}$$

where:

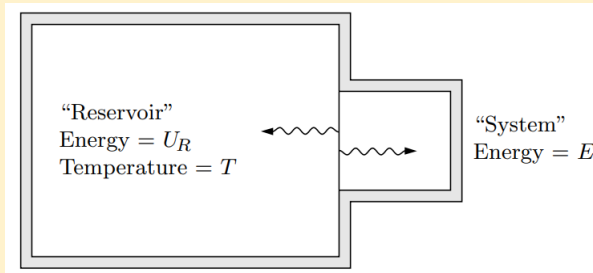
- $p(s)$ is the probability of finding a system in state s

“...the most useful formula in all of statistical mechanics. Memorize it.”

General shape of the Boltzmann Distribution:

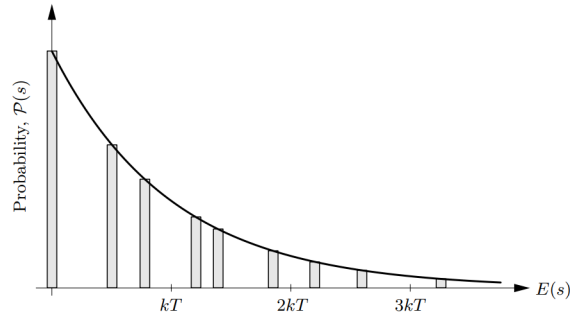
- $1/Z$ is a constant of proportionality to convert the Boltzmann Factor to a probability
- $E(s)$ is the energy of the state s
- k is the [Boltzmann Constant](#)
- T is temperature

Model:



This assumes that $dV_R \approx 0$, but in cases such as the hydrogen atom with infinite energy levels, this is not the case. The new Boltzmann factor would be

$$e^{-(E(s) + PV)/kT}$$



Note:

- $e^{-E(s)/kT}$ is called the Boltzmann Factor
- High energy states are much less likely
- A lower T leads to a sharper falloff of probabilities (horizontally squeeze the graph)
- A higher T leads to a more gradual falloff of probabilities (horizontally stretch the graph)

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6.1 Partition Function

$$Z = \sum_{s=0}^{\infty} e^{-E(s)/kT}$$

= *sum of all Boltzmann factors*

where:

- Z is the partition function
- $E(s)$ is the energy of the state s
- k is the [Boltzmann Constant](#)
- T is temperature

This is derived from enforcing that total probability always sums to 1 and rearranging [6.1 Boltzmann / Canonical Distribution](#).

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6.2 Average Values

(Note that from here on out, $\beta = 1/kT$ will be used more often.)

In general, any average value can be computed as so:

$$\bar{X} = \sum_s X(s)P(s)$$

where:

- $\bar{X}$ is the average of some property
- $X(s)$ is some property of the system with the state s
- $P(s)$ is the probability of that property with the state s

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6.2 Average Energy

$$\overline{E} = \sum_{s=0}^{\infty} E(s)p(s) = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

where:

- $E(s)$ is the energy for a state s
- $p(s)$ is the probability for a state s
- Z is the partition function
- $\beta = 1/kT$

The total energy is just the number of particles times the average energy per particle:

$$U = N\overline{E}$$

A useful derivation:

$$\overline{E^2} - (\overline{E})^2 = -\frac{\partial \overline{E}}{\partial \beta}$$

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6.2 Partition Function of Paramagnets

$$Z = 2 \cosh \beta \mu B$$

where:

- $\beta = 1/kT$
- μ is the magnitude of the magnetic dipole moment
- B is the magnitude of the magnetic field

This equation is derived from the partition function general equation and the energies of the paramagnet listed way above.

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6.2 Rotation of Diatomic Molecules

Number of rotational degenerate states:

$$2j + 1$$

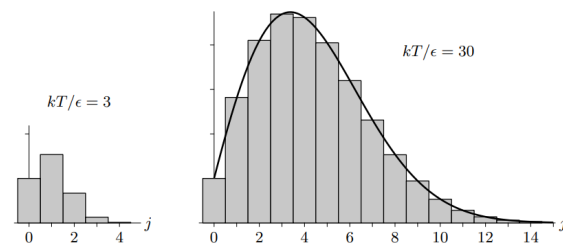
Energy of rotational states (molecules of **non-identical** atoms):

$$E(j) = j(j + 1)\epsilon$$

Partition function (high temperature limit, molecules of **non-identical** atoms):

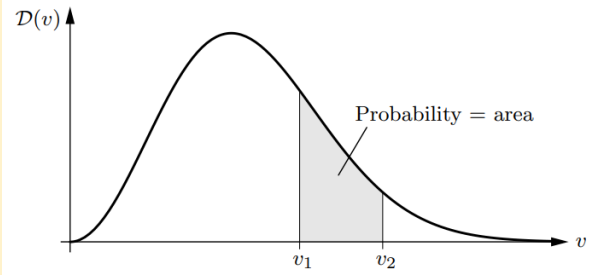
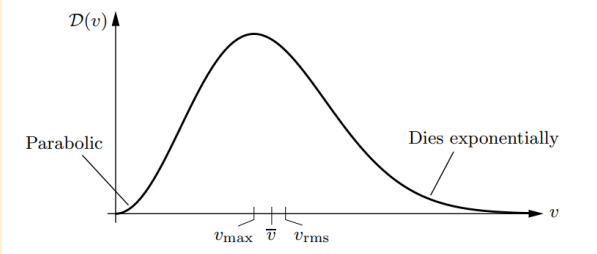
$$Z_{rot} \approx \frac{kT}{\epsilon}$$

Partition function (high temperature limit, molecules of **identical** atoms):



where:

- $j = 0, 1, 2, \dots$ is the energy level
- ϵ is an energy constant that is inversely proportional to the molecule's moment of inertia
- k is the [Boltzmann Constant](#)
- T is the temperature

$Z_{rot} \approx \frac{kT}{2\epsilon}$ Average rotational energy (for molecules with identical or non-identical atoms): $\overline{E}_{rot} = kT$	Both Z_{rot} are obtained by approximating the original Z sigma as an integral, which requires a high temperature. Pages 243-245 of PDF
6.3 Gaussian Integral $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$ An integral of this form is quite common in physics. Note that the antiderivative of the integrand is non-elementary, but using certain techniques, one can evaluate the definite integral over all real numbers. Page 249 of PDF	6.4 Probability in the Maxwell Speed Distribution $Prob(v_1 \dots v_2) = \int_{v_1}^{v_2} D(v) dv$ $Prob(v_1 \dots v + dv) = D(v) dv$  Page 251 of PDF
6.4 Maxwell Distribution $D(v) = \left(\frac{m}{2\pi kT}\right)^{3/2} \cdot 4\pi v^2 \cdot e^{-mv^2/2kT}$ From the equipartition theorem: $v_{rms} = \sqrt{3kT/m}$ By setting the first derivative to zero: $v_{max} = \sqrt{2kT/m}$ By integrating $\int_0^{\infty} vD(v)dv$: $\overline{v} = \sqrt{8kT/\pi m}$	 $4\pi v^2$ is the number of degenerate states for kinetic energy. where: $D(v)$ is the Maxwell Distribution function with units of $(m/s)^{-1}$ m is mass

Due to the coefficients, this is always true:

$$v_{max} < \bar{v} < v_{rms}$$

- k is the [Boltzmann Constant](#)
- T is temperature
- v is speed

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6.5 Partition Function and Free Energy

$$F = -kT \ln Z$$

where:

- F is Helmholtz free energy
- k is the [Boltzmann Constant](#)
- T is the temperature
- Z is the partition function

Using the partial derivative expressions for F , we can find the pressure, chemical potential, and entropy of a system of fixed reservoir temperature just using its partition function.

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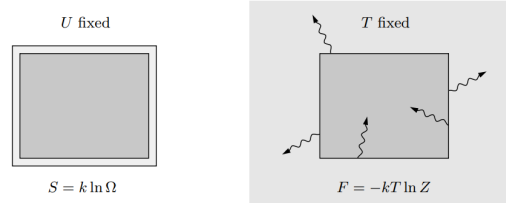


Figure 6.14. For an isolated system (left), S tends to increase. For a system at constant temperature (right), F tends to decrease. Like S , F can be written as the logarithm of a statistical quantity, in this case Z .

Entropy can be thought of as:

$$S = \frac{\bar{E} - F}{T}$$

where $\bar{E}$ is used interchangeably with U (?).

6.6 Composite System Partition Functions

Noninteracting, distinguishable particles:

$$Z_{total} = Z_1 Z_2 \dots Z_N$$

Noninteracting, indistinguishable particles:

$$Z_{total} = \frac{1}{N!} Z_1^N$$

The system must not be too dense for the indistinguishable particles case.

- N is the number of particles

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6.7 Some Quantum Mechanics

Particle in a 1D box:

$$\lambda_n = \frac{2L}{n}$$

$$p_n = \frac{h}{\lambda_n}$$

$$E_n = \frac{p_n^2}{2m} = \frac{h^2 n^2}{8mL^2}$$

Quantum length:

$$l_Q \equiv \frac{h}{\sqrt{2\pi m k T}}$$

Quantum volume:

$$v_Q = l_Q^3$$

- λ is wavelength
- L is the length of the box

	• n is an integer 1, 2, 3, . . . • p is momentum • h is Planck's Constant • E is energy • m is mass • k is the Boltzmann Constant • T is temperature Page 261, 262 of PDF
6.7 Ideal Gas Partition Function For indistinguishable particles: $Z_1 = Z_{tr} Z_{int}$ $Z_{1d} \equiv \frac{L}{l_q}$ $Z = \frac{1}{N!} \left(\frac{V Z_{int}}{v_q} \right)^N$ $\ln Z \approx N[\ln V + \ln Z_{int} - \ln N - \ln v_q + 1]$ where: • Z_{tr} is the translational partition function • Z_{int} is the internal partition function (out of scope of this class) • L is the length of the box • l_q is the quantum length • N is the number of particles • V is volume • v_q is the quantum volume Page 260, 262, 263 of PDF	6.7 Ideal Gas Predictions $U = U_{int} + \frac{3}{2} NkT$ Constant volume heat capacity: $C_V = \frac{\partial U_{int}}{\partial T} + \frac{3}{2} Nk$ Entropy: $S = Nk \left[\ln \left(\frac{V}{N v_q} \right) + \frac{5}{2} \right] - \frac{\partial F_{int}}{\partial T}$ Chemical potential: $\mu = -kT \ln \left(\frac{V Z_{int}}{N v_q} \right)$ where: • U_{int} is the internal energy • N is the number of particles • k is Boltzmann Constant • T is temperature • V is volume • v_q is the quantum volume • F_{int} is the internal Helmholtz free energy • Z_{int} is the internal partition function (out of scope of this class) Page 263, 364 of PDF

Chapter 7

7.1 Gibbs Factor, Grand Partition Function, and Probability

Gibbs Factor:

$$e^{-[E(s) - \mu N(s)]/kT}$$

Grand Partition Function / Gibbs sum:

$$Z = \sum_s e^{-[E(s) - \mu N(s)]/kT}$$

Probability of a state:

$$p(s) = \frac{1}{Z} e^{-[E(s) - \mu N(s)]/kT}$$

where:

- $E(s)$ is the energy of state s
- μ is chemical potential
- $N(s)$ is the number of particles in state s
- k is [Boltzmann Constant](#)
- T is temperature

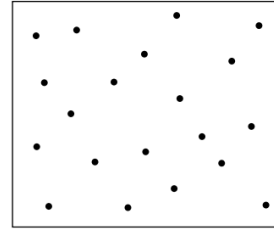
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7.2 Boltzmann vs Quantum Statistics

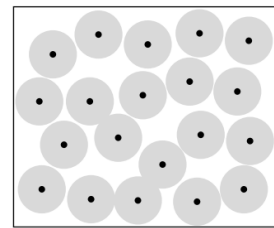
Boltzmann statistics works accurately when

$$Z_1 \gg N$$

so that there is a very low chance of multiple particles occupying the same state.



Normal gas, $V/N \gg v_Q$



Quantum gas, $V/N \approx v_Q$

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7.2 Single-Particle State Probability

$$P(n) = \frac{1}{Z} e^{-n(\epsilon - \mu)/kT}$$

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where:

- Z is the grand partition function
- n is the number of particles
- ϵ is the energy of the single-particle state when it is occupied by *one* particle
- k is [Boltzmann Constant](#)
- T is temperature

7.2 Fermion Grand Partition Function

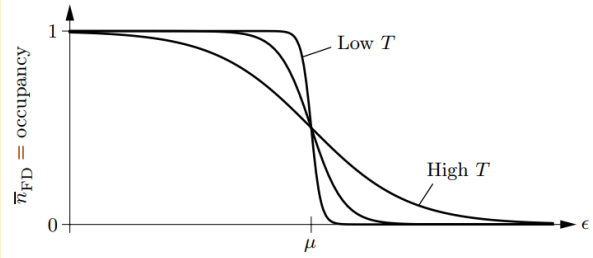
$$Z = 1 + e^{-(\epsilon - \mu)/kT}$$

where:

7.2 Occupancy of a State

$$\bar{n} = - \frac{1}{Z} \frac{\partial Z}{\partial x}$$

$$x \equiv (\epsilon - \mu)/kT$$

• ϵ is a unit of energy (representing a single state) • μ is chemical potential • k is Boltzmann Constant • T is temperature $E(s) = n\epsilon$ Only 0 or 1 fermions can be in this specific state, so the partition function is just these two terms. (One single-particle state) Page 275 of PDF	where: • Z is the grand partition function • ϵ is a unit of energy (representing a single state) • μ is chemical potential • k is Boltzmann Constant • T is temperature Works for both bosons and fermions. Page 276 of PDF
7.2 Fermi-Dirac Distribution $\bar{n}_{FD} = \frac{1}{e^{(\epsilon-\mu)/kT} + 1}$ where: • ϵ is a unit of energy (representing a single state) • μ is chemical potential • k is Boltzmann Constant • T is temperature Applies to fermions only. Derived via $\sum n p(n)$ for $n = 0, 1$.	 Note: • When $\epsilon \gg \mu$, $\bar{n}_{FD} \rightarrow 0$ (most states with energy above μ go unoccupied) • When $\epsilon \ll \mu$, $\bar{n}_{FD} \rightarrow 1$ (most states with energy below μ are occupied) Page 276 of PDF
7.2 Boson Grand Partition Function $Z = \frac{1}{1 - e^{-(\epsilon-\mu)/kT}}$ where: • ϵ is a unit of energy (representing a single state) • μ is chemical potential • k is Boltzmann Constant • T is temperature Derived via the power series convergence of	7.2 Boltzmann Average Occupancy $\bar{n}_{Boltzmann} = e^{-(\epsilon-\mu)/kT}$ where: • ϵ is a unit of energy (representing a single state) • μ is chemical potential • k is Boltzmann Constant • T is temperature Page 277 of PDF

$1/(1 - x)$. For the series to converge, the following *must* be true: $\epsilon > \mu$

$$\begin{aligned} Z &= 1 + e^{-(\epsilon-\mu)/kT} + e^{-2(\epsilon-\mu)/kT} + \dots \\ &= 1 + x + x^2 + \dots \\ &= \frac{1}{1-x}, \text{ where } x = e^{-(\epsilon-\mu)/kT} \end{aligned}$$

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To find the average occupancy in general, use the magical formula

$$\bar{n} = -\frac{1}{Z} \frac{\partial Z}{\partial x}$$

where:

- Z is the grand partition function
- $x = (\epsilon - \mu)/kT$

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7.2 Bose-Einstein Distribution

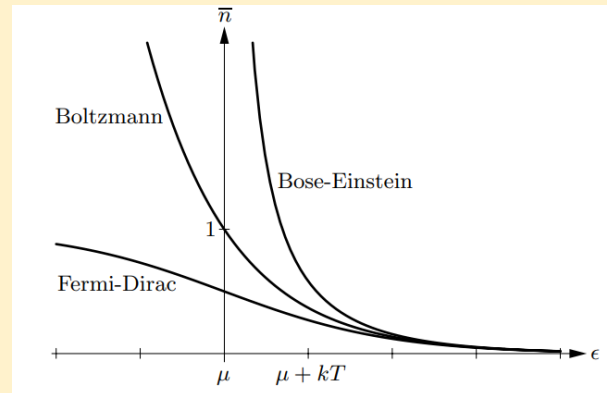
$$\bar{n}_{BE} = \frac{1}{e^{(\epsilon-\mu)/kT} - 1}$$

where:

- ϵ is a unit of energy (representing a single state)
- μ is chemical potential
- k is [Boltzmann Constant](#)
- T is temperature

Note:

- When $\epsilon \gg \mu$, $\bar{n}_{BE} \rightarrow 0$
- When $\epsilon \rightarrow \mu$, $\bar{n}_{BE} \rightarrow \infty$ when ϵ CANNOT be less than μ



Note:

- All three distributions become approximately **equal** when $\epsilon \gg \mu$, so the other two distributions reduce to the Boltzmann distribution at high energies.

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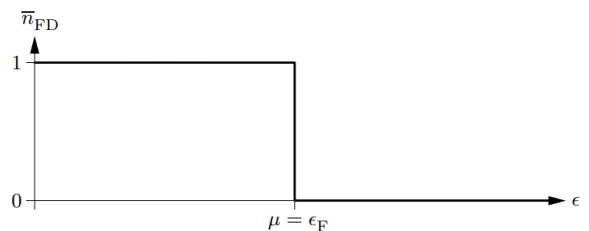
7.3 Degenerate Fermi Gas

$$\epsilon_F \equiv \mu(T = 0)$$

$$\epsilon_F = \frac{h^2}{8m} \left(\frac{3N}{\pi V} \right)^{2/3}$$

Fermi energy is an **intensive** quantity!

The graph of the fermi distribution at $T = 0$ is a step function:



where:

- ϵ_F is Fermi energy
- h is [Planck's Constant](#)
- m is mass
- N is particle number
- V is volume

The derivation follows from using the particle in a 1D box model and finding its appropriate λ , p , and E .

Total energy of a degenerate Fermi gas:

$$U = \frac{3}{5} N \epsilon_F$$

Fermi temperature is not always practical (since the material would melt and boil well before you reach it), but the following is a way to find it:

$$T_F \equiv \epsilon_F / k$$

where k is the [Boltzmann Constant](#).

A gas behaves quantumly if

$$V/N \ll v_Q \text{ or } kT \ll \epsilon_F$$

(both are equivalent)

Degeneracy pressure:

$$P = \frac{2}{3} \frac{U}{V}$$

Bulk modulus:

$$B = \frac{10}{9} \frac{U}{V}$$

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7.3 Boltzmann vs Degenerate Fermi Gas

For the degenerate Fermi gas model to be accurate, any of these must be true (all are equivalent):

$$kT \ll \epsilon_F \text{ or } T \ll T_F \text{ or } T \approx 0$$

In other words:

- Thermal energy is much lower than Fermi energy
- Temperature is much lower than the Fermi temperature
- The gas is “cold”

Even though we may be at room temperature $T = 300K$, this is “basically 0” compared to a very high Fermi temperature.

For the classical (**Boltzmann** statistics) model to be accurate, the other extreme must be true:

$$kT \gg \epsilon_F \text{ or } T \gg T_F$$

In other words:

- Thermal energy is much higher than Fermi energy
- Temperature is much tighter than the Fermi temperature

If NEITHER of these inequalities is satisfied, i.e.

$$kT \approx \epsilon_F \text{ or } T \approx T_F \text{ (roughly)}$$

then, **neither** model is accurate.

Problem 7.20

7.4 Planck Distribution

$$\bar{n}_{Planck} = \frac{1}{e^{hf/kT} - 1}$$

where:

- h is [Planck's Constant](#)
- f is frequency
- k is [Boltzmann Constant](#)
- T is temperature

In the 1D particle in a box model, modes of energies $hf \gg kT$ are negligible.

Since photons are bosons, the Planck Distribution reduces to the Bose-Einstein Distribution, which means:

$$\epsilon = hf$$

$$\mu_\gamma = 0$$

Also, remember that photon energy is given by pc instead of $p^2/2m$ (because they are massless).

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7.4 Energy Density of Photons

Also known as the spectrum of the photons:

$$u(\epsilon) = \frac{8\pi}{(hc)^3} \frac{\epsilon^3}{e^{\epsilon/kT} - 1}$$

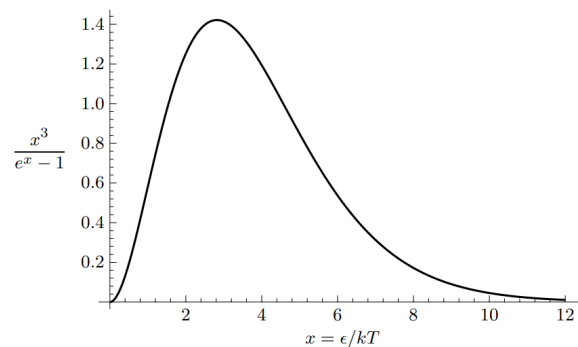
where:

- h is [Planck's Constant](#)
- c is the [Speed of Light](#)
- ϵ is photon energy (hf)
- k is [Boltzmann Constant](#)
- T is temperature

Integrating $u(\epsilon)$ from ϵ_1 to ϵ_2 gives the energy density (energy per unit volume) within that range of energies.

$$\frac{U}{V} = \int_{\epsilon_1}^{\epsilon_2} u(\epsilon) d\epsilon$$

The substitution $x = \epsilon/kT$ is handy.



Wien's Law states that the higher the temperature, the higher the photon energy (lower wavelength and higher frequency).

The spectrum peaks at $\epsilon = 2.82kT$ or $\lambda = hc/4.97kT$.

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7.4 Total Energy Density of All Frequencies

$$\frac{U}{V} = \frac{8\pi^5}{15(hc)^3} (kT)^4$$

where:

- h is [Planck's Constant](#)
- c is the [Speed of Light](#)
- k is [Boltzmann Constant](#)
- T is temperature

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7.4 Heat Capacity, Entropy of a Photon

$$C_V = \frac{32\pi^5 V}{15(hc)^3} (kT)^3$$

$$S(T) = \frac{32\pi^5 V}{45} \left(\frac{kT}{hc} \right)^3 k$$

where:

- V is volume
- h is [Planck's Constant](#)
- c is the [Speed of Light](#)
- k is [Boltzmann Constant](#)
- T is temperature

Class 5/1/2025 (R)

End of the R750:315 Curriculum